

AAKRITI AGGARWAL

SENIOR AI ENGINEER

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AI/ML professional with nearly 4 years of experience in designing, developing, and deploying advanced artificial intelligence and machine learning solutions. Seeking a challenging role in a forward-thinking organization to leverage data-driven insights, drive innovation, and deliver scalable AI solutions that create measurable business impact.

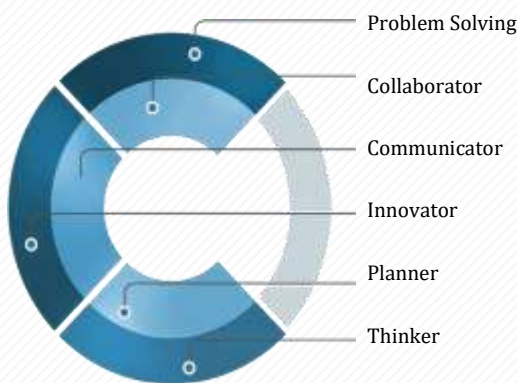


PROFILE SUMMARY

- Proficient in **end-to-end ML lifecycle** from data collection and experimentation to API integration and model deployment.
- An effective communicator** with proven skills in working with cross-cultural teams during the project phase; guiding & training team members and enabling knowledge sharing among the team.
- Adept at collaborating with **cross-functional teams** to translate business needs into robust AI architectures and measurable outcomes.
- Demonstrated thought leadership** through open-source contributions, technical blogging, and participation in AI research and innovation.
- Recognized for **delivering high business impact through AI-driven efficiency improvements**, automation solutions, and customer-facing innovations.



SOFT SKILLS



TECHNICAL SKILLS

- AI & Generative AI:** GenAI (GPT, Claude, Llama), Fine-tuning, SLM Fine-tuning, Prompt Engineering, RAG Systems, Agentic Frameworks, Transformers
- AI Security & Compliance:** Policy Decision Points (PDPs), Prompt Injection & Jailbreak Protection, Data Privacy & PII Masking (DLP Rules), OAuth2/RBAC Access Control, IAM
- Programming & Languages:** Python, C++, High Level System Design
- AI/ML Libraries & Tools:** PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib, Hugging Face, LangChain
- MLOps & Deployment:** MLflow, Docker, GitHub Actions, CI/CD, Model Lifecycle Management, Distributed Inference
- Cloud:** AWS (Lambda, Event bridge, Bedrock, CloudTrail)
- Databases:** Vector Databases, Elasticsearch, Milvus, DynamoDB, S3
- Web & API Frameworks:** FastAPI, Flask, RESTful APIs



AWARDS

- Microsoft MVP (AI) 2025-26:** Recognized for outstanding contributions to AI and the developer community.
- Certified Technical AI Speaker 2025:** Delivered 30+ presentations at top industry conferences and forums.
- LinkedIn Top Data Science Voice 2025:** Acknowledged for thought leadership in AI and data science.
- IBM Watson Q3 Growth Award 2024:** Honoured for driving cultural initiatives for German clients and business impact.
- Published Author 2024:** 80+ technical blogs on GenAI, LLMs, RAG, and enterprise AI on Medium.
- AWS Data Exchange Hackathon 2022:** Secured 4th place for innovative solutions.
- Hackatra Hackathon 2022:** Awarded 1st prize for development and execution excellence.
- IEEE & ICSC Conference Publication 2020:** Research paper on DeepFake Detection methodology.



WORK EXPERIENCE

Jan 2024-Present: IBM Watsonx, Bengaluru as Senior AI Engineer - Band 7B

Key Result Areas:

- Architected and deployed scalable Generative AI solutions across multimodal, RAG, and agentic pipelines using WatsonX and open-source LLMs, drove €6M in client revenue.
- Built an enterprise-scale code-RAG system processing 40+ GB of data, enabling intelligent code-level QA, enhancing developer productivity, and reducing reliance on senior engineers.
- Re-engineered document enrichment pipelines integrated with Watson Discovery, cutting latency from 3000s to 290s (90% reduction) through async parallelization, intelligent caching, and optimized I/O operations.
- Implemented a Policy Decision Point (PDP) within the AI proxy layer to enforce governance and security, enabling policy-based request validation, prompt-injection defense, and enterprise-grade AI compliance.
- Developed advanced multi-agent systems for procurement negotiation, supplier management, and complex enterprise workflows using watsonx Orchestrate, ODM and Coupa.
- Optimized RAG systems with token-aware truncation and dynamic context window management to efficiently handle large-scale enterprise documents.
- Created APIs, microservices, and integration layers to embed AI capabilities into enterprise applications and platforms.

Jul 2022-Jan 2024: SAP Labs, Bengaluru as Software Developer Associate in AI

Feb 2022-Jul 2022: SAP Labs as Intern

Key Result Areas:

- Spearheaded the design and deployment of an LLM-driven testing automation platform for log analysis, bug tracking, and reporting, slashing manual effort by 80%, recognized by MD of SAP Labs India - Sindhu Gangadharan.
- Engineered a proof-of-concept for migrating on-premise systems to a multi-cloud SAP architecture, leveraging LLMs to accelerate development timelines by 60%.
- Scaled infrastructure for 25,000+ enterprise customers through AI-enabled cloud migration and performance optimization initiatives.
- Demonstrated the solution to SAP clients, highlighting measurable improvements in cloud adoption and enterprise AI integration.

Projects & Research

Open Source Contributions

Multi-Agent Generator – Python Library | Creator & Maintainer

- Developed a Python library that translates plain English instructions into fully configured multi-agent AI teams.
- Implemented LLM-agnostic support using LiteLLM for compatibility with GPT, Claude, Llama, and other models.
- Enabled 6+ agentic frameworks for flexible multi-agent architecture design.
- Integrated automatic code generation and workflow visualization for streamlined development.
- Achieved 4,000+ downloads within six months and earned 100+ GitHub stars, reflecting strong community adoption.
- Published on PyPI for easy installation and distribution.

Research Publications

2020 – IsSwap Deep Fake Detection | IEEE | ICSC | Lead Author and Presenter

- Developed a robust video analysis pipeline to detect deepfakes across 400+ samples.
- Applied facial expression analysis using XceptionNet and ResNet50 architectures by using hyperparameter tuning.
- Presented research at IEEE and ICSC conferences, showcasing novel methodologies for multimedia deepfake identification.

Enterprise AI Systems

2025 – Educational RAG Platform for CBSE Schools | AI Architecture Design & Implementation

- Designed and deployed RAG system for an educational NGO, serving 1,000+ students across 4 schools.
- Architected end-to-end RAG system optimized for CBSE curriculum content.
- Developed intelligent retrieval mechanisms to provide students and teachers instant access to contextual learning materials.



EDUCATION

- **Bachelor of Technology (Computer Science)** from Jaypee Institute of Information Technology, Noida (2018-2022)



Technical Content & Blogging

- Published 80+ articles on Generative AI, LLMs, RAG systems, multi-agent frameworks, and enterprise AI solutions.
- Covered GenAI fundamentals, LLM fine-tuning, prompt engineering, system design, AI governance, and cloud deployment.
- Targeted content for AI practitioners, data scientists, and enterprise architects seeking practical AI implementation guidance.



Personal Projects

Autonomous AI Security Auditor

Built an autonomous AI security auditor for AWS Bedrock, leveraging Bedrock Agents, Lambda orchestration, EventBridge-driven scheduled scans, and CloudTrail analysis to detect policy misconfigurations, insecure prompts, data-leak risks, and adversarial vulnerabilities across GenAI pipelines. Implemented S3-based audit logging, IAM configuration analysis, and DynamoDB-backed model-policy mapping to ensure enterprise-grade compliance, traceability, and safety.

IAM Framework for Agentic Systems (In Progress)

Currently developing an IAM framework for LLM-based agent systems on AWS, adding agent identities, RBAC-style tool permissions, policy enforcement (PDP/PEP), and Bedrock Agent integration to enable secure, permission-scoped multi-agent orchestration

AI Evaluation System – IIT Bombay Mentorship

Mentoring an IIT Bombay student to build a hierarchical AI-agent evaluation system where agents self-assess and escalate low-quality outputs. Designed a modular evaluation pipeline using CrewAI and FastAPI, improved reliability over traditional assessment methods.